G H PATEL COLLEGE OF ENGINEERING & TECHNOLOGY

(THE CHARUTAR VIDYA MANDAL (CVM) UNIVERSITY)

DEPARTMENT OF INFORMATION TECHNOLOGY

Multi-Modal RAG Agent for Enterprise Guidelines

INDUSTRIAL INTERNSHIP (202000801)

SECOND PRESENTATION

27TH FEBRUARY, 2026

PRESENTED BY

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EXTERNAL GUIDE

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INTERNAL GUIDE

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CP DEPARTMENT, GCET,

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Company Profile

- **Company Name:** Aziro Technologies Pvt. Ltd.
- **Location:** Pune, Maharashtra

Aziro Technologies is a technology-driven software company focused on:

- AI & Data Science solutions
- Enterprise software development
- Research-oriented problem solving

Strong emphasis on:

- Modern AI stacks
- Scalable architectures
- Real-world problem solving

Internship environment encouraged:

- Independent exploration
- Hands-on learning
- Research-based implementation

WHAT IS RAG?

Retrieval-Augmented Generation (RAG) : Retrieval-Augmented Generation (RAG) is an AI framework that improves LLM accuracy by fetching relevant, up-to-date information from external, trusted knowledge bases, such as internal company documents or databases, before generating a response.

Traditional LLM:

- Static knowledge
- Cannot access enterprise documents
- Risk of hallucination

RAG:

- Retrieves relevant document chunks
- Injects context into LLM
- Generates grounded answers

- It is more accurate, reduces hallucination and, is more enterprise-ready.

System Overview

1. **Vision Extraction**
   Extracting information from documents
2. **Chunking**
   Dividing text into manageable sections
3. **Embeddings**
   Creating numerical representations of text
4. **Vector Database**
   Storing and organizing embeddings efficiently
5. **Hybrid Retrieval**
   Combining different search techniques
6. **LLM Answer**
   Generating responses using language models
7. **Feedback Loop**
   Iterating process based on results
8. **Final Output**
   Compiling the results into a final document

VISION-BASED PDF EXTRACTION

Vision-Based Ingestion:

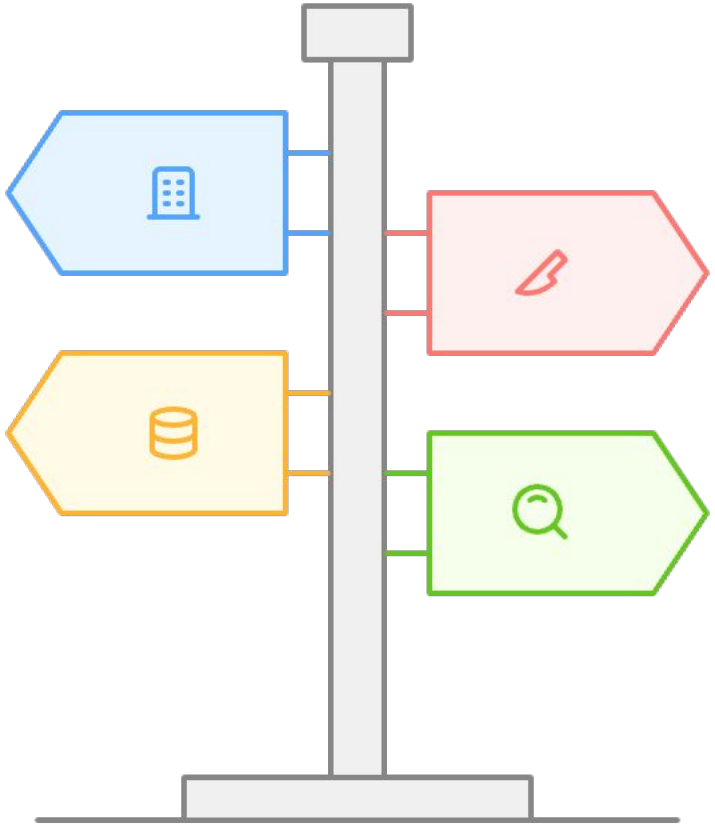
Instead of basic PDF parsing:

- Each page converted to image (PyMuPDF)
- Sent to Groq Vision model
- Reconstructed as structured Markdown

Advantages:

- Handles complex tables
- Extracts text from screenshots
- Preserves layout
- Works for scanned PDFs

Core RAG Pipeline Architecture



Structured Extraction
Enforced header hierarchy and context-aware subheaders

Embeddings & Vector Storage
Uses all-MiniLM-L6-v2 and Chroma database

Intelligent Chunking
Header-based semantic splitting and structural continuity

Hybrid Retrieval
Combines dense retrieval and BM25 scoring

Screenshots

Ingest Chat Debug Extraction

Ingest Documents

Upload PDFs

Drag and drop files here
Limit 200MB per file • PDF

IBMMaaS360_Best_Practices_for_Policies.pdf 429.0KB

IBMMaaS360_Technical_Support_Guide.pdf 0.6MB

Process & Embed

Processing 2 file(s) — skipping already-ingested...

RAG Pipeline:

Ingest Chat Debug Extraction

Ask a question:

What are the recommendations under Best Practice #4?

Generate Answer

Answer

Best Practice #4 – Recommendations for Restricting Device Features

- Limit access to Siri when the device is locked.
- Restrict Bluetooth use (or make Bluetooth non-discoverable).
- Disable mock-location capabilities.
- Prevent syncing documents to iCloud (while still allowing other iCloud backups and Photo Stream).
- Block the camera, screen-capture functions, and YouTube when required by industry regulations.
- For iOS Safari, apply the following settings:
  - Keep fraud warnings enabled.
  - Block pop-ups.
  - Accept cookies only from sites the user has visited.

Sources used: [IBMMaaS360_Best_Practices_for_Policies.pdf, Page 8]

Conclusion

Vision-based extraction
Utilizes visual data for effective information extraction

Hybrid retrieval
Combines various data retrieval methods

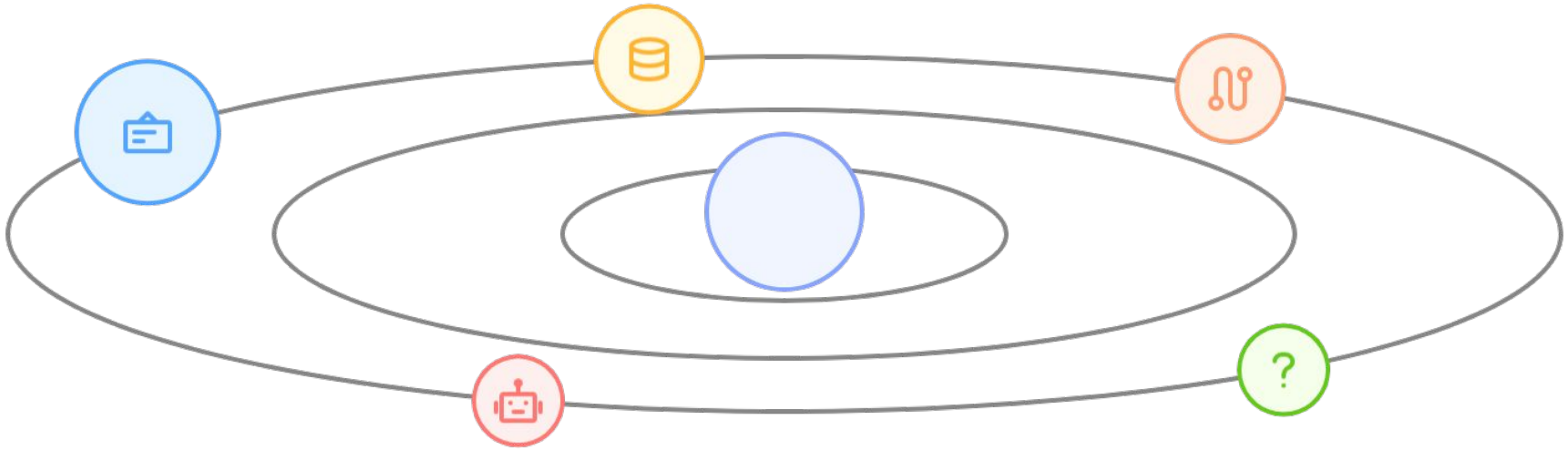
Multi-chunk reasoning
Processes multiple information segments simultaneously

Future Scope

Problem Statement
Formulate a structured problem statement for RAG system

Knowledge Base
Build a verified medical knowledge base

Evidence-Based Responses
Ensure grounded, evidence-based responses



RAG Chatbot
Develop a chatbot focused on 4-5 cancers

Patient Questions
Enable patients to ask about their conditions

References

- Aziro Technologies – Official Website - https://www.aziro.com/en
- LangChain Documentation - https://docs.langchain.com/
- Docling Documentation - https://docling-project.github.io/docling/

THANK YOU…